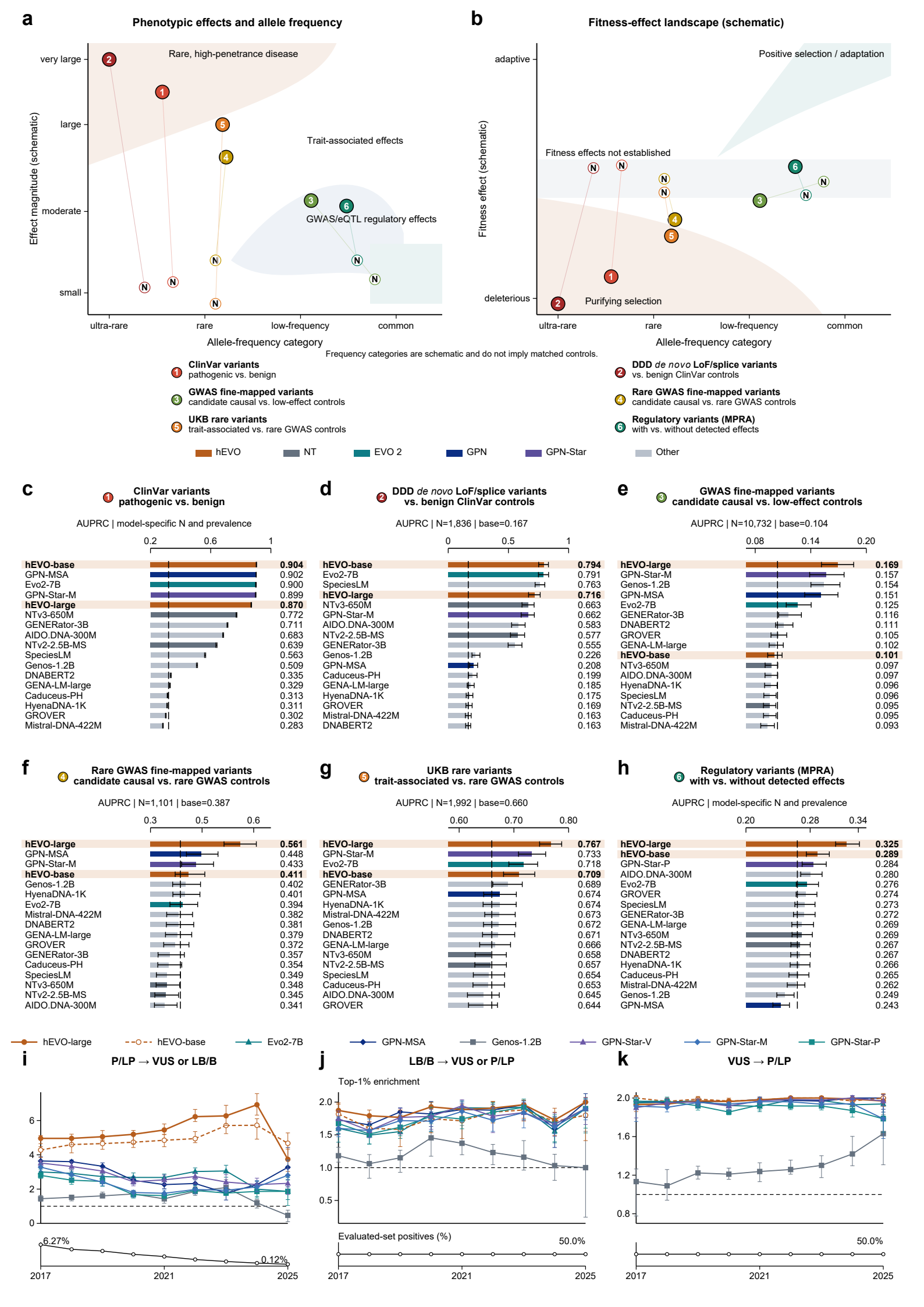

a
Phenotypic effects and allele frequency
very large
large
moderate
small
Effect magnitude (schematic)
Rare, high-penetrance disease
Trait-associated effects
GWAS/eQTL regulatory effects
ultra-rare
rare
low-frequency
common
Allele-frequency category
b
Fitness-effect landscape (schematic)
adaptive
deleterious
Fitness effect (schematic)
Positive selection / adaptation
Fitness effects not established
Purifying selection
ultra-rare
rare
low-frequency
common
Allele-frequency category
Frequency categories are schematic and do not imply matched controls.
1 ClinVar variants
pathogenic vs. benign
2 DDD de novo LoF/splice variants
vs. benign ClinVar controls
3 GWAS fine-mapped variants
candidate causal vs. low-effect controls
4 Rare GWAS fine-mapped variants
candidate causal vs. rare GWAS controls
5 UKB rare variants
trait-associated vs. rare GWAS controls
6 Regulatory variants (MPRA)
with vs. without detected effects
hEVO
NT
EVO 2
GPN
GPN-Star
Other
c
1 ClinVar variants
pathogenic vs. benign
AUPRC | model-specific N and prevalence
0.2
0.6
1
hEVO-base 0.904
GPN-MSA 0.902
Evo2-7B 0.900
GPN-Star-M 0.899
hEVO-large 0.870
NTv3-650M 0.772
GENERator-3B 0.711
AIDO.DNA-300M 0.683
NTv2-2.5B-MS 0.639
SpeciesLM 0.563
Genos-1.2B 0.509
DNABERT2 0.335
GENA-LM-large 0.329
Caduceus-PH 0.313
HyenaDNA-1K 0.311
GROVER 0.302
Mistral-DNA-422M 0.283
d
2 DDD de novo LoF/splice variants
vs. benign ClinVar controls
AUPRC | N=1,836 | base=0.167
0
0.5
1
hEVO-base 0.794
Evo2-7B 0.791
SpeciesLM 0.763
hEVO-large 0.716
NTv3-650M 0.663
GPN-Star-M 0.662
AIDO.DNA-300M 0.583
NTv2-2.5B-MS 0.577
GENERator-3B 0.555
Genos-1.2B 0.226
GPN-MSA 0.208
Caduceus-PH 0.199
GENA-LM-large 0.185
HyenaDNA-1K 0.175
GROVER 0.169
Mistral-DNA-422M 0.163
DNABERT2 0.163
e
3 GWAS fine-mapped variants
candidate causal vs. low-effect controls
AUPRC | N=10,732 | base=0.104
0.08
0.14
0.20
hEVO-large 0.169
GPN-Star-M 0.157
Genos-1.2B 0.154
GPN-MSA 0.151
Evo2-7B 0.125
GENERator-3B 0.116
DNABERT2 0.111
GROVER 0.105
GENA-LM-large 0.102
hEVO-base 0.101
NTv3-650M 0.097
AIDO.DNA-300M 0.097
HyenaDNA-1K 0.096
SpeciesLM 0.096
NTv2-2.5B-MS 0.095
Caduceus-PH 0.095
Mistral-DNA-422M 0.093
f
4 Rare GWAS fine-mapped variants
candidate causal vs. rare GWAS controls
AUPRC | N=1,101 | base=0.387
0.3
0.5
0.6
hEVO-large 0.561
GPN-MSA 0.448
GPN-Star-M 0.433
hEVO-base 0.411
Genos-1.2B 0.402
HyenaDNA-1K 0.401
Evo2-7B 0.394
Mistral-DNA-422M 0.382
DNABERT2 0.381
GENA-LM-large 0.379
GROVER 0.372
GENERator-3B 0.357
Caduceus-PH 0.354
SpeciesLM 0.349
NTv3-650M 0.348
NTv2-2.5B-MS 0.345
AIDO.DNA-300M 0.341
g
5 UKB rare variants
trait-associated vs. rare GWAS controls
AUPRC | N=1,992 | base=0.660
0.60
0.70
0.80
hEVO-large 0.767
GPN-Star-M 0.733
Evo2-7B 0.718
hEVO-base 0.709
GENERator-3B 0.689
GPN-MSA 0.674
HyenaDNA-1K 0.674
Mistral-DNA-422M 0.673
Genos-1.2B 0.672
DNABERT2 0.671
GENA-LM-large 0.666
NTv3-650M 0.658
NTv2-2.5B-MS 0.657
SpeciesLM 0.654
Caduceus-PH 0.653
AIDO.DNA-300M 0.645
GROVER 0.644
h
6 Regulatory variants (MPRA)
with vs. without detected effects
AUPRC | model-specific N and prevalence
0.20
0.28
0.34
hEVO-large 0.325
hEVO-base 0.289
GPN-Star-P 0.284
AIDO.DNA-300M 0.280
Evo2-7B 0.276
GROVER 0.274
SpeciesLM 0.273
GENERator-3B 0.272
GENA-LM-large 0.269
NTv3-650M 0.269
NTv2-2.5B-MS 0.267
DNABERT2 0.267
HyenaDNA-1K 0.266
Caduceus-PH 0.265
Mistral-DNA-422M 0.262
Genos-1.2B 0.249
GPN-MSA 0.243
hEVO-large
hEVO-base
Evo2-7B
GPN-MSA
Genos-1.2B
GPN-Star-V
GPN-Star-M
GPN-Star-P
i
P/LP → VUS or LB/B
6
4
2
0
6.27%
0.12%
2017
2021
2025
j
LB/B → VUS or P/LP
Top-1% enrichment
2.0
1.5
1.0
0.5
Evaluated-set positives (%)
50.0%
2017
2021
2025
k
VUS → P/LP
2.0
1.6
1.2
0.8
50.0%
2017
2021
2025